Natural Selection Study Guide

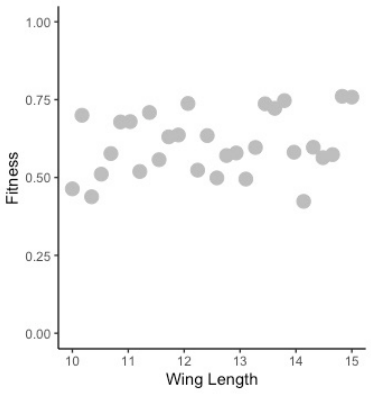
This study guide provides examples for the topics that you might see on the exam over natural selection. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture, the lecture slides and video on Canvas, and your homework questions to review all the topics that have been covered.

1. Which of the scenarios below would be considered macroevolution?
    a. A population of antelope that lives at high elevation for several generations experiences an increase in the frequency of an allele that contributes to increased oxygen levels in the bloodstream
    b. A population of songbirds colonizes a remote island system and over several million years diversifies into a dozen different species specialized on different food sources
    c. A population of mosquitoes is exposed to an insecticide and over a few generations, develops partial resistance to the insecticide
    d. In a population of lizards, two body colors occur, black and light tan. In response to climate warming in recent decades, the light tan coloration has increased in frequency in the population

2. In class, we discussed the frequency-dependent selection example of body color in female dragonflies. In this species, males prefer to mate with the females with the andromoph body color when they are less than half of the population; however, when andromorph body color is present in more than half the female population, males prefer to mate with females of the gynomorph body color. What factor(s) drive(s) body color selection in this system? *Select all that apply*
    a. Rarity
    b. Camouflage
    c. Male mate preference
    d. Female mate preference
    e. Exposure to predators

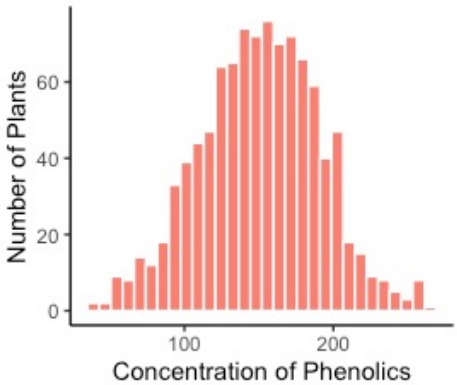
3. After tracking a population of birds for several years, you plot the relationship between wing length and fitness, as shown in the figure below. This figure represents _[blank1]_ selection and the next generation of birds should have _[blank2]_ wing lengths compared to the current generation.

Options for Blank 1: directional, stabilizing, disruptive, no
Options for Blank 2: larger, smaller, similar



4. Which of the following are examples of **balancing selection**? *Select all that apply*
    a. Small plants have higher fitness in summer and large plants have higher fitness in winter, resulting in a population that has both small and large plants
    b. When birds lay too many eggs, they can't provide for them all, but when they lay too few they are more likely to lose them all to predation, resulting in medium-sized clutches
    c. Green morphs of an insect species have higher fitness on leaves while brown morphs have higher fitness on bark and branches. Both persist in the population.
    d. The establishment of a new highway causes higher fitness for birds with shorter wings because they can veer away from oncoming cars

5. Many plants produce chemical compounds that protect them from predators, either because they make the plants unpalatable (taste bad) or toxic. You hypothesize that a newly invading insect species will cause selection for higher levels of phenolics, a particular type of defensive compound. You measure plants in a meadow that has not yet been exposed the invasive insect and plot the levels of phenolic compounds in the histogram below. Which of the necessary and sufficient conditions for natural selection does this data demonstrate?
    a. Variation
    b. Heritability
    c. Differential Success
    d. All of them
    e. None of them



6. You are in charge of a strawberry breeding program and are trying to use artificial selection to breed larger strawberries. Your most recent crop has an average berry size of 11.5 grams. You choose a subset of plants that make the largest berries to breed for the next generation. The average berry size for this subset is 12.8 grams. After breeding, the next generation has an average berry size of 12.2 grams.

    a. What is the response to selection in this system?

    b. What is the heritability of this trait?

7. What is the difference between a pattern of selection and a driver of selection?

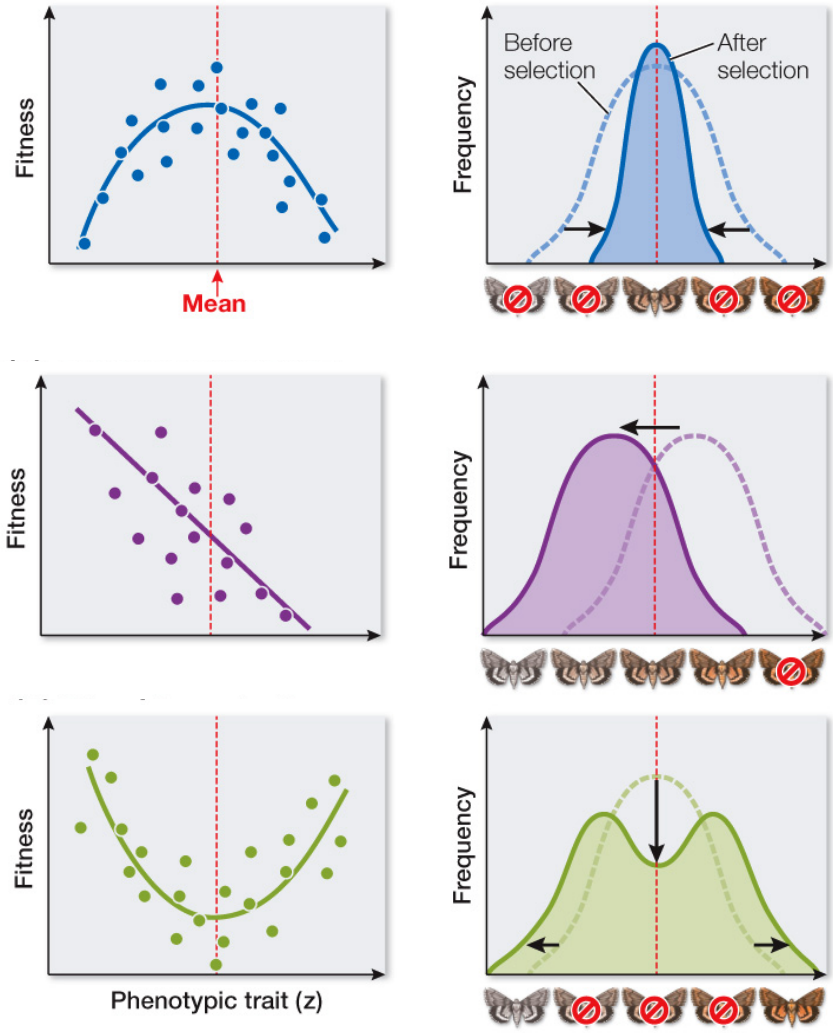
8. What are the necessary and sufficient conditions a population must meet in order for natural selection to occur?

9. How is frequency-dependent selection similar to and different from directional selection?

10. Under what conditions would altruism be supported by natural selection?

11. What types of factors constrain the process of natural selection? Phrased another way, why does natural selection not always result in organisms that are 'perfectly' adapted to their environment?

12. Describe in words the relationships shown in the graphs at below. Label each pair as illustrating directional selection, stabilizing selection, or disruptive selection.



13. Bacteria are asexual organisms, and a bacterium reproduces by duplicating its DNA and splitting into two separate cells. If a single bacterial cell that is sensitive to vancomycin (an antibiotic) is placed in an environment that contains vancomycin, it will die. Now consider another single bacterial cell, also sensitive to vancomycin, that is allowed to divide for many generations into a larger population. If this population is placed into vancomycin-containing environment, some bacteria will grow.
    a. Why do you see growth in the larger population (that is allowed to divide), but not with the single cell?

    b. Which of the populations meets the conditions for natural selection? What type of selection is occurring?

14. You are studying whether harvesting large fish from a population will select for smaller body size in the fish. You create an experimental population of fish, all the same age. The average body size of your experimental population is 100 grams. You then harvest the largest fish, removing them from the population and preventing them from reproducing. The average body size of the

remaining fish is 75 grams. The fish remaining are all allowed to breed. What is the predicted body size for their offspring if heritability (h^2) of body size is…..

a. 1
b. 0.5
c. 0